

Question_5:

Code:

```
1  /*
2  Question No: 5
3  Given the beautiful code (Prüfer sequence) of a tree,
4  construct the original tree.
5  Use the smallest numbered leaf at every step.
6  */
7  #include <iostream>
8  #include <vector>
9  #include <queue>
10 using namespace std;
11
12 int main(){
13     int n;
14     cin>>n;
15
16     vector<int> a(n-2),deg(n+1,1);
17     for(int &x:a){
18         cin>>x;
19         deg[x]++;
20     }
21     priority_queue<int,vector<int>,greater<int>> q;
22
23     for(int i=1;i<=n;i++)
24         if(deg[i]==1)
25             q.push(i);
26     for(int x:a){
27         int leaf=q.top();
28         q.pop();
29
30         cout<<leaf<<" "<<x<<"\n";
31
32         if(--deg[x]==1)
33             q.push(x);
34     }
35     int a1=q.top(); q.pop();
36     int a2=q.top();
37     cout<<a1<<" "<<a2<<"\n";
38 }
```

Output:

Test Case_1:

```
5
2 2 4
1 2
3 2
2 4
4 5
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5
3 1 2
4 3
3 1
1 2
2 5
PS C:\Users\Prabin Yadav\
```